

SALAJ CHOUDHARY

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SRM Institute of Science and Technology, Trichy, India

EDUCATION

SRM Institute of Science and Technology, Trichy	2023 – 2027
B.Tech in Information Technology	CGPA: 9.33 / 10
Subodh Public School, Jaipur	2022
Class XII (CBSE – PCM)	70%
Ryan International School, Jaipur	2020
Class X (CBSE)	89%

SKILLS AND INTERESTS

Programming Languages: Java, Python, JavaScript, C, C++
Frameworks & Technologies: React.js, React Native, Node.js, Spring Boot, HTML, CSS, Tailwind CSS, Bootstrap
Databases: MySQL, MongoDB, Firebase Realtime Database
Machine Learning: scikit-learn (Isolation Forest, Random Forest), Pandas, NumPy
Tools: Git, GitHub, AWS, Expo, Vite

PROJECTS

Smart Asset Tracking System	2025
- Engineered a real-time asset tracking system leveraging smartphones as IoT nodes, eliminating dependency on dedicated GPS hardware - Developed a cross-platform mobile application with continuous background location tracking optimized for low power consumption - Architected a low-latency (< 500 ms) data synchronization pipeline using Firebase Realtime Database and event-driven updates - Designed a web-based dashboard for multi-asset monitoring with real-time route visualization - Applied machine learning models (Isolation Forest, Random Forest) achieving > 85% anomaly detection accuracy - Implemented geofencing algorithms and automated alerting mechanisms for anomaly detection	
Data Wiping and Secure Erasure Tool	Aug 2025 – Present
- Developed a secure data sanitization system ensuring irreversible deletion of sensitive data from storage devices - Implemented multi-pass overwrite algorithms compliant with DoD 5220.22-M and Gutmann standards - Built a React-based dashboard for real-time monitoring and control of wipe operations - Optimized performance using multithreading and concurrent I/O handling - Integrated logging and audit mechanisms for traceability and compliance - Improved efficiency by approximately 60% through automation	
Student Database Management System	Feb 2025 – Apr 2025
- Designed and implemented a CRUD-based academic management system for structured and unstructured data handling - Integrated MySQL and MongoDB for hybrid database architecture and optimized querying - Developed authentication and role-based access control for secure system usage - Implemented filtering, sorting, and indexed search to improve query performance - Built a Java Swing-based UI following modular design principles	
NGO Website Development (Naya Sawera NGO)	Jun 2025 – Present
- Developing a responsive web application to enhance NGO outreach and engagement - Implementing secure donation workflows, volunteer registration, and form handling - Built reusable UI components with validation, modals, and dark mode support - Optimized accessibility, performance, and cross-browser compatibility - Collaborating using Git-based version control and iterative development practices	
2D Arcade Game (Flappy Bird-Inspired)	Jul 2024 – Oct 2024
- Developed a 2D arcade game using Java with real-time rendering and event-driven architecture - Implemented physics logic including gravity, velocity, and collision detection algorithms - Engineered a high-frequency game loop for smooth rendering and responsive input handling - Designed modular object-oriented components for game state, rendering, and physics engine	